Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

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1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with a parity-based shooting procedure. The goals are to validate the method on systems with known analytic solutions, demonstrate numerical convergence and accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Extensions to scattering problems are also explored as a secondary component.

2 Computational approach

In dimensionless units, the Schrödinger equation is written as

$$\psi''(x) = 2\,[V(x) - E]\psi(x).$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For symmetric potentials, even and odd states are treated separately using parity conditions at $x = 0$. Eigenvalues are found by scanning for sign changes in the boundary mismatch and then refining them with bisection. The correctness of the numerical approach is verified through comparison with analytical solutions where available and through systematic convergence studies.

The overall numerical workflow is implemented in a modular way in the accompanying Python code. The core components include a Numerov integrator for propagating the wavefunction, a shooting module that enforces parity-based boundary conditions and constructs the mismatch function, and analysis utilities for convergence studies and parameter sweeps. This separation of concerns allows each numerical component to be validated independently and makes the implementation easier to interpret.

2.1 Shooting and root finding

The boundary-value eigenproblem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a given trial energy E, the Schrödinger equation is integrated and a boundary mismatch is evaluated. For symmetric potentials we exploit parity and define

$$\text{even states:} \quad M(E) = \psi'_E(0), \tag{1}$$
$$\text{odd states:} \quad M(E) = \psi_E(0). \tag{2}$$

Eigenvalues correspond to the roots of $M(E) = 0$, which are located by bracketing sign changes and refined with bisection. To make the procedure transparent, we plot $M(E)$ versus E and mark the bisection iterates; zero crossings identify the allowed energies.

In the implementation, the mismatch function is evaluated numerically using the propagated wavefunction. For even states, the derivative at the origin is computed using a high-order finite-difference stencil to ensure that the overall accuracy of the method is not limited by the boundary evaluation. This detail is crucial for achieving consistent convergence behavior in the harmonic oscillator case.

3 Results

Each potential is used to validate, test, or demonstrate different aspects of the numerical method, ranging from analytic benchmarks to nontrivial quantum behavior.

3.1 Infinite square well

The infinite square well serves as a benchmark problem with known analytic spectrum

$$E_n = \frac{(n+1)^2\pi^2}{8a^2}.$$

The potential is

$$V(x) = \begin{cases} 0 & |x| \leq a, \\ \infty & |x| > a \end{cases} \qquad \text{(implemented numerically with a large finite } V_0\text{)}.$$

The numerical eigenvalues agree with the analytic values to high precision. Convergence with respect to grid spacing is demonstrated with log–log plots, and the fitted slopes indicate power-law behavior $\Delta E \propto h^p$.

3.2 Harmonic oscillator

For the harmonic oscillator with $V(x) = \frac{1}{2}\omega^2 x^2$, the numerical spectrum reproduces the exact levels $E_n = \omega(n+\frac{1}{2})$. Since the problem is defined on an unbounded domain, we truncate to $[-x_{\max}, x_{\max}]$ and impose decay conditions at the boundaries. To avoid contamination by the exponentially growing solution, we use inward shooting: the solution is initialized in the asymptotic region where it decays and integrated toward the origin. The resulting wavefunctions exhibit the correct parity and node structure without spurious growth near the edges.

A subtle numerical issue arises when enforcing the even-parity condition $\psi'(0) = 0$. Although the Numerov method itself is high-order, the accuracy of the eigenvalues can be degraded if the derivative at the origin is evaluated with a low-order finite-difference approximation. In this implementation, a fourth-order one-sided stencil is used when sufficient grid points are available, ensuring that the boundary condition is enforced with comparable accuracy to the integration scheme. This correction was essential for obtaining consistent agreement with alternative methods such as Runge–Kutta integration, and illustrates how boundary-condition accuracy can limit an otherwise high-order method.

3.3 Double-well potential

The quartic double-well potential is used to demonstrate nontrivial quantum behavior beyond analytic benchmarks and produces pairs of nearly degenerate states. The lowest even and odd states correspond to symmetric and antisymmetric combinations of localized wavefunctions. The energy splitting between these states is analyzed as a function of the barrier height, illustrating quantum tunneling effects.

3.4 Finite square well

The finite square well demonstrates the flexibility of the solver. Unlike the infinite case, only a finite number of bound states exist, depending on the well depth. The numerical results capture this behavior correctly.

3.5 Scattering and resonant tunneling

In addition to bound states, we extend the solver to study one-dimensional scattering from finite potentials. For a given energy E, the wavefunction is propagated across the domain and decomposed into incoming and reflected components on the left and a transmitted component on the right.

The transmission and reflection coefficients $T(E)$ and $R(E)$ are computed from the amplitudes of these components and satisfy the conservation relation

$$T(E) + R(E) \approx 1.$$

For a single finite rectangular barrier, the numerical transmission is compared to the known analytical result, providing an additional validation of the method.

For a double-barrier potential, the transmission spectrum exhibits sharp resonance peaks corresponding to quasi-bound states in the central well. These resonances are clearly visible in $T(E)$ and illustrate quantum tunneling and interference effects.

3.6 Convergence study

We analyze numerical errors by varying both the grid spacing h and the domain size $x_{\max}$.

Grid convergence. For fixed $x_{\max}$, the energy error follows approximately

$$\Delta E \propto h^p,$$

where p is estimated from log–log fits and reported in the figure legends. The observed values $p \approx$ 3–4 are consistent with high-order Numerov accuracy, with mild degradation due to boundary truncation and root-finding tolerances.

Dependence on domain size. For problems on unbounded domains (e.g., the harmonic oscillator), truncation at finite $x_{\max}$ introduces boundary error. For small $x_{\max}$ the wavefunction is artificially cut off, producing large errors. As $x_{\max}$ increases, the error decreases until it reaches a plateau set by discretization, root-finding tolerance, and floating-point precision. Thus this plot measures sensitivity to domain size rather than pure discretization convergence.

Convergence is primarily established using systems with known analytic solutions (square well and harmonic oscillator), and similar stable behavior is observed for more complex potentials such as the double well, although quantitative error estimates are less accessible there.

A comparison with a fourth-order Runge–Kutta (RK4) integrator was also performed for the harmonic oscillator. Both methods solve the same shooting problem, but RK4 evolves the wavefunction and its derivative simultaneously, whereas Numerov exploits the specific structure of the Schrödinger equation. Initially, RK4 appeared more accurate due to a lower-order derivative approximation in the Numerov boundary condition. After correcting this issue, both methods exhibit comparable convergence behavior, with Numerov achieving similar or better accuracy for the same grid spacing. This highlights the importance of treating boundary conditions with the same order of accuracy as the integration scheme.

Representative plots include wavefunctions, probability densities, root-finding diagnostics, convergence curves, and energy splitting as a function of potential parameters.

4 Discussion

The correctness of the code is verified through multiple checks. Analytical benchmarks confirm the accuracy of the computed eigenvalues, while normalization ensures that the wavefunctions remain physically meaningful. Convergence studies provide strong evidence that the numerical results are stable with respect to discretization parameters.

The double-well analysis highlights a key physical phenomenon: tunneling-induced energy splitting. As the barrier height increases, the splitting decreases, reflecting reduced overlap between the localized states. This demonstrates that the numerical method captures not only correct energies but also meaningful physical behavior.

The results also illustrate typical numerical considerations, such as sensitivity to grid resolution and domain size, and emphasize the importance of matching the accuracy of boundary conditions to that of the integration scheme.

The agreement with analytical results, together with consistent convergence behavior and well-behaved root-finding diagnostics, demonstrates the correctness and robustness of the numerical implementation. Differences in observed convergence rates across states are attributed to boundary truncation and solver tolerances rather than defects in the underlying method.

5 Conclusion

The Numerov method, combined with parity-based shooting and robust root-finding, provides an efficient and accurate tool for one-dimensional bound-state problems, with controlled errors from discretization, domain truncation, and floating-point precision. In addition to reproducing known spectra, it captures physically interesting effects such as tunneling-induced energy splitting in a double well.

The extension to scattering problems further demonstrates the flexibility of the approach as a secondary component, allowing the study of transmission, reflection, and resonant tunneling phenomena within the same numerical framework.